Ziyan(Cecilia) Xia

Foster City, CA, 94404 | xiaziyan1999@gmail.com | xiaziyan1999 | GitHub | github.io

SUMMARY

Product Data Scientist with senior-level expertise building end-to-end analytics and modeling systems for B2B SaaS products. Designed production pipelines and dashboards that reduced decision latency by 32% and delivered over $75 M in customer cost savings via tire-health insights. Partnered with product, engineering, and go-to-market teams to translate large-scale data into actionable product and business recommendations. Aiming to leverage data-driven solutions to accelerate product impact and drive growth.

PROFESSIONAL EXPERIENCE

Revvo Technologies Inc. (AI Road Safety Platform) **Jul 2022 - Dec 2025**
Data Analyst/Analytic Engineer (Full-Time) *San Mateo, CA*

- **Metric Design & Product Insights:** Designed end-to-end KPI frameworks for a large-scale IoT ecosystem (1B+ miles, 5K+ tire events), defining self-serve real-time metrics and customer-facing insights that enabled proactive road-safety alerts and contributed $75M+ in customer operating cost savings.
- **Experiment Design, A/B Testing & Causal Inference:** Designed and ran field A/B tests (Python, Git) evaluating firmware updates, gateway repositioning, and signal-filtering algorithms; compared telemetry distributions and downstream model outputs to ensure backward compatibility and prevent production regressions at scale.
- **Statistical Forecasting & Machine Learning:** Built and deployed production regression, classification, clustering, and survival models for tire risk prediction, sensor positioning, and depot detection, achieving up to 90% accuracy across 50+ enterprise fleets in 25+ U.S. states.
- **Cross-Functional Impact & Stakeholder Management:** Partnered with Product, Engineering, and GTM teams (Customer Success, Sales, Marketing) to define success metrics, evaluate feature launches and account health, and deliver recommendations that guided product iterations, ML model calibration, LLM agent evaluation, and C-suite decisions
- **Anomaly Detection & Root Cause Analysis:** Automated investigation of abnormal sensor behavior with Python/SQL diagnostics and cross-device comparisons, distinguishing environmental noise from hardware issues and reducing false-positive tire replacements by 80%+ (200+/week → <40/week).
- **Production Analytics Workflows:** Engineered end-to-end analytics pipelines (Python, SQL/GCP, Java, Looker) for ML-driven insights in Revvo's TireIQ platform; used Git, unit testing, and CI/CD for reliable deployments, cutting investigation time 75% (2 hrs → 30 min).
- **Data Modeling & ETL Pipelines:** Designed scalable data models and production ETL pipelines for structured and unstructured telemetry; implemented Python/SQL data validation and quality monitoring to ensure reliable downstream analytics.
- **Production-grade Algorithm Design & Optimization:** Designed and optimized sensor auto-registration algorithms using statistical methods and GPS telemetry, correcting noisy mappings and false positives, reducing installation calibration time 85% (5 days → 18 hrs).

University of California, San Francisco - Roland Henry Lab **Feb 2020 - Feb 2021**
Undergraduate Research Assistant (Part-Time) through UC Berkeley URAP *San Francisco, CA*

- **Time-Series Forecasting & Predictive Analytics:** Built end-to-end Python analytics and modeling pipelines on high-frequency wearable (Fitbit) clinical data, using ARIMA-regression hybrid models to forecast mobility trends and behavioral patterns across longitudinal patient cohorts and identify meaningful changes in mobility over time.
- **Statistical Analysis & Data Quality:** Established subject-level baselines and deviation signals through within-subject analysis and hypothesis testing in Python, validated data quality, and identified anomalous observations, which improved detection of meaningful motor-function changes over time
- **Clinical Data Processing & Feature Engineering:** Cleaned and transformed longitudinal clinical data from disparate sources, implementing missing-value imputation, noise reduction, validation rules, and feature extraction to create reliable analytical datasets and stable inputs for downstream predictive modeling.
- **Exploratory Analytics & Stakeholder-Ready Insights:** Conducted exploratory analysis of clinical time-series data to uncover behavioral patterns, seasonal effects, and activity cycles, translating findings into interpretable analyses that supported hypothesis generation, model design, and understanding of longitudinal patient outcomes

EDUCATION

Carnegie Mellon University **Aug 2021 - May 2022**
M.S., Statistics *Pittsburgh, PA*
- **GPA:** 3.95/4.0

Central China Normal University **Sep 2017 - Jun 2021**
B.S., Statistics *Wuhan, China*
- **GPA:** 87.69/100

University of California, Berkeley **Aug 2019 - May 2020**
Visiting Student (BISP), Statistics *Berkeley, CA*
- **GPA:** 3.5/4.0

SKILLS

- **Analytics & Experimentation**: A/B Testing, Experiment Design, Causal Inference, Metric Design, Time Series, LLM, NLP
- **Programming**: Python (NumPy, Pandas, scikit-learn, statsmodels, TensorFlow, PyTorch), SQL, Java, R, JavaScript, Bash
- **Data Engineering**: BigQuery/PostgreSQL, Spark, Airflow, ETL Pipeline, Batch Processing, Data Modeling, API Integration
- **Data Visualization Tools**: Looker, Tableau, Python Plotly, Power BI, R Shiny App, Excel/Google Sheets
- **Production Practices**: Git (Version Control), Docker, CI/CD, Unit Testing, Agile/Scrum (Jira)